Term 6 – Heating and Cooling

This is a checklist of all of content from the **heating and cooling topic** that you could be assessed on in **term 6.**

Remember the strategy - **Identify** a weakness --> **Learn** it --> **Test** it.

Start by RAGing your understanding for each of the outcomes below. Then focus on learning each of the **red** and **amber** outcomes before **testing yourself.**

Topic	Learning Outcomes	RAG Yourself	
		1	2
Heating and Cooling	Recall the three states of matter and the four state changes		
	State the boiling and freezing point of water		
	Explain what is meant by expansion and contraction		
	Describe what happens to the particles in a substance when it is heated		
	Describe how thermal energy transfers between hotter and colder objects		
	Give the standard units of temperature and energy		
	Compare temperature to thermal energy in terms of kinetic energy of particles		
	Describe how thermal energy transfer occurs through solids through conduction		
	Define the terms insulator and conductor with examples of each		
	Describe how thermal energy transfers through liquids and gases by convection		
	Describe how surfaces and temperature affect thermal radiation transfer		
	State three methods used to reduce heat transfer in a house		
	Explain insulation methods in terms of reducing conduction, convection or thermal radiation		